Imperial College London

Assignment Feedback

Group Number:	Group 7
Project Name:	Board Game
Assignment:	Final Report

Comments from Marker 1:

- Abstract doesn't really explain why the project exists, but does provide details on the design and the evaluation.
- Introduction reads more like an abstract and talks a bit too much about the final design.
- Good set of aims and objectives, nicely sets out the magnitude of the project.
- Introduction is a bit waffly - you spend about four lines say you went to John Chilton School to get feedback (try to be more efficient in your writing).
- Good matrix comparing the potential input devices some images to better demonstrate what they are.
- Try breaking your background section into packets or sub-sections to make it more readable.
- Good set of user requirements, Technical requirements have some good quantifiable values but would possibly have been better as a table.
- Good looking design, well shown with images that are labelled really well. Nice use of flow charts to explain function.
- Testing section is generally pretty good but could be improved by sectioning the methods and results for each test.
- An honest evaluation of the function well presented in the evaluation matrix, along with a good list of future amendments to improve the design.
- Interesting approach to demonstrating the workload distribution, good breakdown of tasks.
- Risk Assessment looks robust, possibly you don’t need to add a preventative measure if the RPN is so low. Might question the scorings in some areas (particularly the Occurrence and Detection rankings) in general the severity is low in most cases. Good effort.
- Good work on covering the main ethics issues possible.
- Good breakdown of the manufacture makes it easy for someone to recreate the work, nicely thought through.

Comments from Marker 2:

Abstract
A generally good, clear and complete abstract. You don’t need to discuss the system’s shortcomings in the abstract, though.

Aims & Objectives
Your approach to these is excellent – identifying your overall aim, then breaking it down into stages and sub-goals and so planning your activity. Well done.

Requirement Definition
This is a little verbose and could be tightened up.

Technical Requirements
These are well done – clear, testable, quantitative.

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Final Design

Your description of the design is back to front. You should describe the principles of the game first, and then how you will achieve those principles, and then how you construct or design the various elements which comprise the system. You do describe the principles of the game briefly in the Abstract, but the Abstract is a brief summary of the whole project, and does not in any way replace the need to describe the Design Principles in this section.

Furthermore, there is no description of the evolution of this design, the alternatives available already, the alternative approaches considered and why the present system was decided upon. This is all important detail on the development of the project, which is missing.

In general, your detailed design is not bad. Why have you used stainless steel tube but mild steel connectors? You could have easily gotten away with mild steel tube and connectors and painted it all for corrosion protection.

Since you detect the arrival of the ball through the goal, could you not have used this signal to switch on the ball return mechanism and then switch it off once the ball is returned to play?

4.2 your experiences on re-design and improvement after the Festival show how important realistic testing is – and good that this was included in your evaluation programme

Are your options for increasing the power of the flipper not much greater than you consider? What about altering the mechanical advantage of the solenoid/flipper gearing, or changing the size or power of the solenoid; you do not discuss how you decided the dimensions of these in your initial design, so this looks like unthought design decisions you made.

Likewise, I am surprised that the solenoids were not screwed or bolted to the baseboard (in fact, you don't describe how they were fixed).

Increasing the power of the flippers might also enable you to change to a denser ball (metal?) which would then maintain its momentum better (pinball games use metal balls, generally?)

References

A good range of references are presented, but your bibliographic details are not always complete.

Risk Analysis

A good and thorough Risk Analysis is described.

If you have any concerns or queries regarding the grade your assignment has been awarded then please raise the matter with the course lead – Dr Ian Radcliffe (i.radcliffe@imperial.ac.uk)